Codebusters C

Syosset High School - CSE

July 2025



- You will have **50 minutes** to complete this test.
- You may bring up to **(3)** 4 or 5 function calculators, if you have any additional please place it underneath your seat
- You may use the space on the exam as scratch paper, but please make sure that your answers are clear. Either write your answers separately and box them, or write them directly over/under the ciphertext.
- You may take apart the test as you wish, however you should write your team number **on top of every page** and a **250 point deduction** will be taken if the pages are **not** put back in the correct order
- The first question on the exam is **Timed**. If you believe you have solved the question in the first 10 minutes, signal to the event supervisor who will come over to check it. If you are correct, your time will be recorded and a bonus issued; if you have more than 2 errors you will be notified and allowed to retry for a bonus throughout the first 10 minutes.
- If you successfully complete the timed question after 10 minutes, you may still get the base points for the question however it will not be checked and no bonus will be awarded.
- There are 3 questions marked for **Special Bonus**: 10, 25, 26
- Page 2 contains scoring information. Pages 3 and 4 contain the reference materials. Page 5 (front of the 3rd sheet) is the **Timed Question**.

Team Name: _____

Team Number: _____

Written By:

Grace Lu (Syo '26)

Thanks to Aneesh Iyer (Mason '25) for helping with the format. :)

Scoring Table

Exam Score

Question	Points	Score
1	254	
2	202	
3	231	
4	273	
5	213	
6	332	
7	283	
8	372	
9	292	
10	311	
11	272	
12	454	

Question	Points	Score
13	173	
14	273	
15	204	
16	344	
17	292	
18	153	
19	201	
20	232	
21	441	
22	282	
23	402	
24	323	

Question	Points	Score
25	372	
26	272	
27	182	
28	243	
29	273	
30	211	
31	280	
32	274	
33	264	
Total:	9180	

Timed Question

Question Score: Circle one: 250 0

Time: _____
minutes : seconds

Timed Bonus: $(600 -$ _____ $) \times 2 =$ _____
time in seconds timed bonus

Total TQ Score: _____ $+$ _____ $=$ _____
question score timed bonus total TQ score

Special Bonus

Circle one. Special bonus is only awarded for solving the question for full score. Special bonus questions: 10, 25, 26.

- 1 Correct = 150
- 2 Correct = 400
- 3 Correct = 750
- 0 Correct = 0

Total

Final Score = _____ $+$ _____ $+$ _____ $=$ _____
Exam Score Total TQ Total Special Bonus

Rank: _____

Tables

The tables below may be useful.

Keys	a	b	c	d	e	f	g	h	i	j	k	l	m
A,B	n	o	p	q	r	s	t	u	v	w	x	y	z
C,D	o	p	q	r	s	t	u	v	w	x	y	z	n
E,F	p	q	r	s	t	u	v	w	x	y	z	n	o
G,H	q	r	s	t	u	v	w	x	y	z	n	o	p
I,J	r	s	t	u	v	w	x	y	z	n	o	p	q
K,L	s	t	u	v	w	x	y	z	n	o	p	q	r
M,N	t	u	v	w	x	y	z	n	o	p	q	r	s
O,P	u	v	w	x	y	z	n	o	p	q	r	s	t
Q,R	v	w	x	y	z	n	o	p	q	r	s	t	u
S,T	w	x	y	z	n	o	p	q	r	s	t	u	v
U,V	x	y	z	n	o	p	q	r	s	t	u	v	w
W,X	y	z	n	o	p	q	r	s	t	u	v	w	x
Y,Z	z	n	o	p	q	r	s	t	u	v	w	x	y

A	AAAAA
B	AAAAB
C	AAABA
D	AAABB
E	AABAA
F	AABAB
G	AABBA
H	AABBB
I/J	ABAAA
K	ABAAB
L	ABABA
M	ABABB

N	ABBAA
O	ABBAB
P	ABBBA
Q	ABBBB
R	BAAAA
S	BAAAB
T	BAABA
U/V	BAABB
W	BABAA
X	BABAB
Y	BABBA
Z	BABBB

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

Frequency of English Letters

E - 12.51%	S - 6.54%	C - 3.06%	G - 1.96%	K - 0.67%
T - 9.25%	R - 6.12%	U - 2.71%	W - 1.92%	X - 0.19%
A - 8.04%	H - 5.49%	M - 2.53%	Y - 1.73%	J - 0.16%
O - 7.60%	L - 4.14%	F - 2.30%	B - 1.54%	Q - 0.11%
I - 7.26%	D - 3.99%	P - 2.00%	V - 0.99%	Z - 0.09%
N - 7.09%				

Frequency of Spanish Letters

E - 14.08%	I - 5.98%	M - 3.08%	Y - 1.09%	Z - 0.47%
A - 12.16%	L - 5.24%	P - 2.89%	V - 1.05%	Ñ - 0.17%
O - 9.20%	U - 4.69%	B - 1.49%	G - 1.00%	X - 0.14%
S - 7.20%	D - 4.67%	H - 1.18%	F - 0.69%	K - 0.11%
N - 6.83%	T - 4.60%	Q - 1.11%	J - 0.52%	W - 0.04%
R - 6.41%	C - 3.87%			

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, "burro" is considered as five letters: "b-u-r-r-o" and not as four letters "b-u-rr-o."

Morse Code

A	●−
B	−●●●
C	−●−●
D	−●●
E	●

F	●●−●
G	––●
H	●●●●
I	●●
J	●–––

K	−●−
L	●−●●
M	––
N	−●
O	–––

P	●––●
Q	––●−
R	●−●
S	●●●
T	−

U	●●−
V	●●●−
W	●––
X	−●●−
Y	−●––
Z	––●●

0	–––––
1	●––––

2	●●–––
3	●●●––

4	●●●●−
5	●●●●●

6	−●●●●
7	––●●●

8	–––●●
9	––––●

●	E
●●	I
●●●	S
●●●●	H
−●●●	B
–––––	0
●●●●●	5

−	T
●−	A
●●−	U
●●●−	V
−●●−	X
●––––	1
−●●●●	6

−●	N
●−●	R
●●−●	F
−●−●	C
●●–––	2
––●●●	7

––	M
●––	W
●−●●	L
−●––	Y
●●●––	3
–––●●	8

−●●	D
●––●	P
––●●	Z
●●●●−	4
––––●	9

−●−	K
●–––	J
––●−	Q

––●	G
–––	O

Timed Question. [250 Points] Solve this **Aristocrat**. When you have finished, **raise your hand and say your favorite ice cream flavor** so your answer can be checked and time recorded.

OV NFU UGNDUBUZJ YZLYYJ QON HONQFUM, NFU YZCHUDV WR

NFU YWUZM XZCCBUM PUDJ YOVN LRMUD NFU FCN VLR.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	2	4	4	0	6	1	2	0	3	0	3	4	9	4	1	2	3	0	0	13	4	2	1	6	5
Replacement																										

1. (254 points) Solve this **K2 Aristocrat**

BSP NVIAAHXXW EIA QVXXOJKR, IKO I BHPP QPVV BSHXCRS

BSP MHXUPK EJKOXE.

Replacement																									
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	4	1	0	3	0	0	4	4	2	4	0	1	1	3	6	2	2	4	0	1	4	1	7	0	0

2. (202 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **TIME**.

$$\begin{pmatrix} T & I \\ M & E \end{pmatrix} = \begin{pmatrix} 19 & 8 \\ 12 & 4 \end{pmatrix}$$

SQKSCECWUOOGSCYWKYFYDKOCKUWS

3. (231 points) Decode this phrase that was encoded using the **Baconian** cipher. You are told that the plaintext ends with RE

BABAAAABBBABBAAAABBABAAAAAABBAABBBAABAAABBAAABBABBAABA

AABBAABBABBAABAABBABBAABAAABBBAABAAAABAABAABBAABAAABBAA

BAABAABAAAAABABAABBBAABAAABAABABBAAAABAABABAABAABAAABBB

AAAAABAABAAABBBABAAABAAABAAAABAABAABAAABBAABAAABABBAAAA

ABAAAAABAAABBAAAAABBBAAABBABAAABBABBAABBABABAAAABBAAAAB

AABAABAABAAABBBAABAABAAAAAABAA

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4. (273 points) Solve this **Aristocrat**

LWCVKZ LVJX SJXO AYS LWKROXR NSKXZ, VKR HA'Z ZVMX

AS ZVB MSFXZ LVJX LWKROXRZ SM NSKXZ HK ALXHO

NSRHXZ.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	1	1	0	0	1	0	4	0	3	7	6	3	3	4	0	0	6	8	0	0	6	3	11	1	9
Replacement																										

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5. (213 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher with a keyword of **CHAIR** and a polybius key of **CHANGE**.

45 24 25 59 94 55 24 34 69 55 33 44 34 68 54

24 63 34 69 87 53 26 34 48 55 25 35 25 59 94

55 24 34 69 87 53 26 56 37 55 62 33 57 38 64

43 33 36

	1	2	3	4	5
1					
2					
3					
4					
5					

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6. (332 points) Solve this **K1 Xenocrypt**

PUHMUHM PG F SF NPMXHF, EMGV OFJPF LMTFXPFLF ÑMUHM

FIMG. VJZPFTMUHM, UV TM ÑYXHF ÑMUHM GYPLVXF.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	0	0	0	1	12	4	7	1	2	0	3	13	1	3	1	7	0	0	1	3	6	4	0	4	2	1
Replacement																											

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7. (283 points) Solve this **Aristocrat**

HMU IUGXYQGN RAU HAVQGX HB FURDU RGHRAPHQPR HB XB

HB HMU GBAHM IBFU RGW LQGW NRGHR.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	6	0	1	0	2	8	10	2	0	0	1	3	2	0	2	4	8	0	0	7	1	2	3	1	0
Replacement																										

8. (372 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **OHNO**.

$$\begin{pmatrix} O & H \\ N & O \end{pmatrix} = \begin{pmatrix} 14 & 7 \\ 13 & 14 \end{pmatrix}$$

PUBUVQBLMYZQMOMAMLGAOEMOYZJSIADHDGPUNSDEAODHMWABVAIRENFGVAGRZYCVUR

9. (292 points) Solve this **Aristocrat**

QITBRQ NSJI TBL CTSRB DBKSLB UFWRFMTI, QRW JTB

QHHAJBW TBL KLFBRWJ SK IQXFRM FI.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	9	1	1	0	5	0	2	5	4	3	4	2	1	0	0	5	6	4	6	1	0	4	1	0	0
Replacement																										

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10. (311 points) Solve this **fractionated morse**. ★***This question is a special bonus question.***

R B I R S X S S D O S P N S A T

Z O T B B S I V A R X

S A B M R U I N C G Y N W M I L

W M N Z I N T X A R B C N T X A K V

Replacement																										
	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

11. (272 points) Solve this **Aristocrat**

SLH PGUH AKZ GW SLH SIHH, KWO SLH DKS AKZ TW SLH

EITNWO ALHIH SLH DLGQOIHW ZSTTO.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	0	0	2	1	0	3	11	4	0	4	7	0	1	4	1	1	0	8	4	1	0	5	0	0	3
Replacement																										

12. (454 points) Solve this **K1 Patristocrat**

HSEHL SETBS VREVE BQNDS NPECB EBWSD HSEHL SPSQN

CASBW SKQND SIEAB LKNPE CBBWS IASLF SA

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	8	3	3	9	1	0	4	2	0	2	4	0	5	0	3	3	1	14	1	0	2	3	0	0	0
Replacement																										

13. (173 points) Decode this phrase that was encoded using the **Porta** cipher with a keyword of **GRADES**.

DPRSL POQFV ENQXU SCOQQ YSSWO TBTWN BKGGS NKLFA

LKXZV DURBK GSDJQ DF

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14. (273 points) Decode this phrase that was encoded using the **Baconian** cipher.

-..-...---..-..-...--..-.-.....-....---...-..-...-.--.-

.--..-..-...---..-..-..-..-.....--...-..-....-.-....-..

-.....-.....-.-.-.-..--..-........--.....--..--.-....

..-....-...--..

15. (204 points) Solve this **Aristocrat**

WSTNLKWSUKIHM, HIJNY ISVNWLUJI QINQHI SNK KN HECI

QUYK SESIKM-SESI. AWK NKXIL AWEHPESJ AHNVBY PN SNK.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	1	1	0	5	0	0	6	10	3	9	3	2	10	0	2	3	0	10	1	3	2	5	1	3	0
Replacement																										

16. (344 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **DEAD**.

$$\begin{pmatrix} D & E \\ A & D \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 0 & 3 \end{pmatrix}$$

HVGCNAQHOANGLMJAAFLMYZLFYTLHYJNQONHVQDDQONBYZH

17. (292 points) Decode this phrase that was encoded using the **Checkerboard** cipher. You are told that the plaintext begins with **THEY**.

AE EC RE KR BS RE BC RE BR BA AE EA BR RS RA KS

EE AE RC KS AE EA AE EE BA BA ER RE RR EE EA ER

RE AE EC RE KR BS RE BC RE

18. (153 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher with a keyword of **EINSTEIN** and a polybius key of **THINK**.

69 48 48 69 46 56 47 68 56 56 27 69

25 49 57 25 37 26 28 59 22 37 38 27

68 45 57 65 25 49 54 35 68 38 28 55

55 46 56 39 57 38 28 89 24 69 38 58

	1	2	3	4	5
1					
2					
3					
4					
5					

19. (201 points) Solve this **Aristocrat**

AVU AUNZVUT ATLUE AH MXUNE VLF LRRHZURZU NA AVU

ATLNX, YWA AVU FAWEURAF ELE RHA YUXLUDU VLG.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	12	0	0	1	5	3	1	3	0	0	0	7	1	4	0	0	0	5	0	3	13	6	2	3	2	3
Replacement																										

20. (232 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **EDITOR**.

EEOOA ERMYP TQLTN TDBED PEUOS GTIEV IODHT GLHEO A

RANT OGOTH TCESI

21. (441 points) Solve this **fractionated morse**.

C W P O X M N D M T

Z S M L S O M B J H Y N L T F M C

Y D M T P N E X C G B

T J M I F S O J G

Replacement																									
	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

22. (282 points) Solve this **cryptarithm** for 12435 8769. Write out your final answer and box it.

Base 10 Addition
Answer: 12435 8769 (two words)

```
     HELP
     ONLY
+     EAT
---------
     NOOO
```

23. (402 points) Solve this **K1 Patristocrat**

IZWHP FAWFJ GWVIS LFTIW SRIZW QSPFV LPGVF TWVSJ

IPRVT HBSGG TQCWI SJGW

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	1	1	0	0	5	5	2	6	3	0	2	0	0	0	4	2	2	6	4	0	5	8	0	0	2
Replacement																										

24. (323 points) Solve this **K2 Xenocrypt**

WC QYWFOH WLV EPT IPWLOW VFHZDW, OVFOH KPW FH JPÑH

ÑHLEYL WF VXMHCPOH. VDHLV WMOV EPT ZVFMVÑH.

Replacement																											
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	0	2	2	3	6	0	10	1	1	1	5	3	0	3	6	6	1	0	0	2	0	9	9	1	2	2

25. (372 points) Solve this **K1 Patristocrat** ★***This question is a special bonus question.***

OYVNZ XEOYR OJRNI ZNZSC VEVRG OYVNO GVVOJ RGEVU

PWXGV ROURE XVGRE UOYVQ GVNVE TVPWN ERBVN OYVNE

RBVSZ ODVCR NOJVV B

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	3	2	1	8	0	6	0	1	3	0	0	0	9	11	2	1	10	2	1	3	19	2	3	5	4
Replacement																										

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26. (272 points) Solve this **cryptarithm** for 013678. Write out your final answer and box it. ★***This question is a special bonus question.***

Base 10 Multiplication
Answer: 013678 (not a word)

```
     DONE
x      AN
---------
    ALOED
```

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27. (182 points) Decode this phrase that was encoded using the **Porta** cipher with a keyword of **HUNGER**.

DRXCLV NOMOXL XOLUSL XHIEVP DRXCPF TXHTRV EQNDEP

UWIEDZ

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28. (243 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **LOOKED**.

IEERE UCDYS ROVNC HHAOX SLCDB EDORX COOIE TEMRX H

EKER AEANR SNDED SOSEX

29. (273 points) Decode this phrase that was encoded using the **Checkerboard** cipher. You are told that the plaintext ends with **EAT IT**.

FP OP RA RN GN FL RT GN FA GP RN GN FL OT FP OP

RA RL FA OT GL RA GN FT FP OP FA FP SL FA GP FP

RA FN FP FL RL FA FP RL OP FA GP FN RA FA FP OL

FP

30. (211 points) Decode this phrase that was encoded using the **Porta** cipher with a keyword of **WALKS**.

TJWIK DASZR VRSBK SRYMW CQUSE ABISE ONQRF HGXWU WVI

31. (280 points) Decode this phrase that was encoded using the **Baconian** cipher.

SERGE FETCH NUDIE FACIE JOLTS HEATS RAINY PIETA IXNAY

YOKES ACTIN ISLES ALDER LEASE VELDS ENDER BRAVO FOCAL

BAIRN LEVER VEALS SULFA BERRY KAYAK EXTRA BOXED TANSY

TRAWL VOTER FARTS MELEE SABRE SQUAW SUEDE UNBAN AGONE

ERECT FRIZZ EAVES FLAKY INLAY MAYBE GRAPE RERAN NOBLE

FORTE

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32. (274 points) Solve this **Aristocrat**

OCYNDCT ECYYK AE ASKJDUNZU, NE GY BAB ZJU ECYYK

HYCC NZB RYKU VNCCAZP NECYYK BFDAZP OCNEE.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	4	10	3	7	1	1	1	0	2	5	0	0	7	2	2	0	1	1	1	4	1	0	0	10	5
Replacement																										

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33. (264 points) Solve this **Aristocrat**

UCR TO CF OD OSE? CF TO OSE HFNSYOF CF MDOJ CTO

KSPDBTJF VFRNCSTI SJ ONCDDM RFOJFBESR.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	2	7	5	3	9	0	1	1	4	1	0	2	3	10	1	0	4	7	5	1	1	0	0	1	0
Replacement																										